SIM
(Service Intelligence Matrix)

Decision Support System for Customer Satisfaction Optimization.

Environment：RStudio ChatGPT

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The Challenge —

Engineering Insights from Scale

df	89656 obs. of 83 variables
df_a	7 obs. of 6 variables
df_avg_month	99 obs. of 4 variables
df_b	7 obs. of 6 variables
df_csi	37168 obs. of 84 variables
df_csi_summary	90 obs. of 9 variables
df_long	941088 obs. of 36 variables
df_low	37168 obs. of 84 variables
df_m	89459 obs. of 83 variables
df_model	37168 obs. of 69 variables
df_model_lag	90 obs. of 13 variables
df_model2	90 obs. of 11 variables
df_sa_month	1003 obs. of 4 variables
df_safe	89656 obs. of 83 variables
df_SAsummary	57 obs. of 9 variables
df_scored	37168 obs. of 38 variables
df_text	2651 obs. of 84 variables
df_trend_week	89 obs. of 8 variables

High-Dimensional Data Reshaping Challenges

Data Scale:

37K+ Surveys ➔ **940K+** Observations via R-based reshaping.

Prioritizing Complex Service SOPs

Objective:

To pinpoint "High-Leverage" actions that impact Customer Satisfaction Rating.

Methodology—

Automated Pipeline & Logic

```r
# 2) Unify the time format
parse_dt <- function(x) {
  # If Excel serial number (double) → Convert to POSIXct
  if (is.double(x)) return(as_datetime((x - 25569) * 86400))
  # Turn to Text format
  parse_date_time(as.character(x),
                  orders = c("ymd HMS", "Y/m/d H:M:S", "Y-m-d H:M:S",
                             "mdy HMS", "mdy HM", "ymd HM"))
}
```

```r
df_long <- df_safe %>%
  filter(是否回函 == 1) %>%
  pivot_longer(
    cols = all_of(make.names(question_cols_raw, unique = TRUE)),
    names_to = "問題_safe",
    values_to = "分數"
  ) %>%
  mutate(分數 = as.numeric(分數)) %>%
  filter(!is.na(分數))
df_model <- df_long %>%
  pivot_wider(
    names_from = 問題_safe,
    values_from = 分數,
    values_fn = mean
  ) %>%
  filter(rowSums(!is.na(select(., starts_with("X")))) >= 10)
```

```r
agent_style_matrix <- plot_df_agent %>%
  filter(問題 %in% question_map$raw) %>%
  left_join(question_map, by = c("問題" = "raw")) %>%
  select(服務專員姓名, safe, 重要度) %>%
  pivot_wider(
    names_from  = safe,
    values_from = 重要度
  ) %>%
  column_to_rownames("服務專員姓名")

agent_style_matrix[is.na(agent_style_matrix)] <- 0
```

AI-Powered Technical Workflow

Collaboration Mode: ChatGPT

Leveraged **ChatGPT** to optimize complex R functions.

Key Innovation:

Developed a Universal Date Parsing Function that automatically standardizes inconsistent time formats.

Data Reshaping & Memory Optimization

From Wide to Long:

Implemented pivot_longer logic to expand 37K surveys into a granular 940K+ observation set, enabling per-item correlation analysis.

Key Integrity Control:

Used try-error handling and custom validation scripts to ensure zero data loss during massive transformations.

Statistical & Matrix Logic

The SIM Matrix:

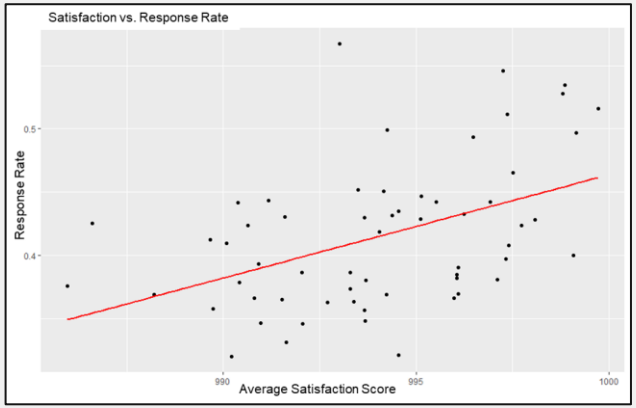
Developed a Service Intelligence Matrix using normalized weights to calculate SA Match Scores, quantifying the alignment between service advisor performance and customer expectations..

Correlation Rigor:

Applied Spearman Rank Correlation to pinpoint non-linear relationships within the observations.

Strategic Insights—

Driving Excellence Through Data



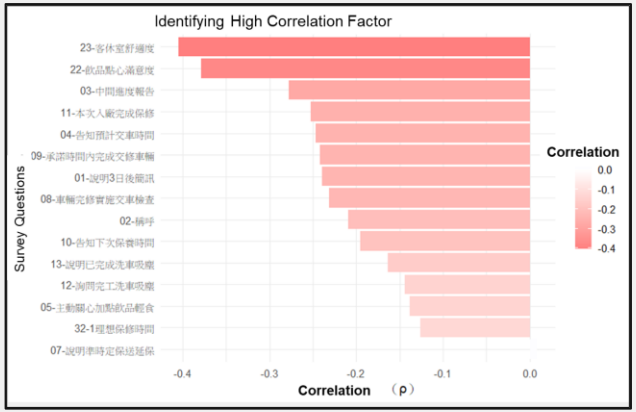
Satisfaction vs. Response Rate

Translation of Findings:

"Correlation Mapping between Satisfaction and Response Engagement."

Strategic Insight:

By using regression analysis to identify core trends and eliminate random noise, stable improvements in satisfaction, rather than short-term data fluctuations.



The Correlation Research

Translation of Findings:

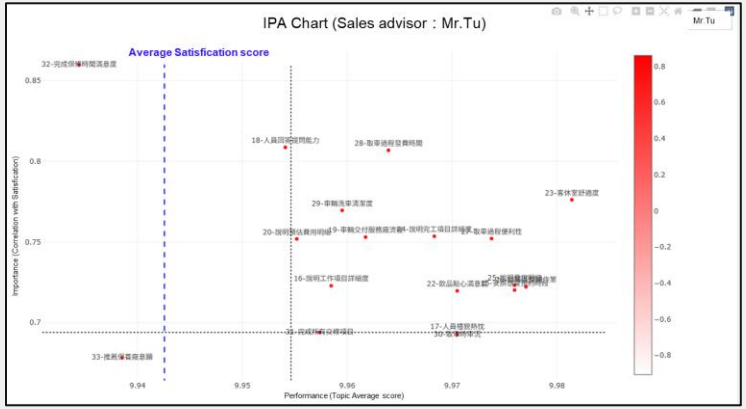
"Key Drivers of Low Satisfaction: A Spearman Correlation Analysis."

Strategic Insight:

The study identified "23 - Guest lounge comfort" as a key factor with the impact on overall satisfaction. This guides on which should be prioritized for optimization.

Strategic Insights—

Driving Excellence Through Data



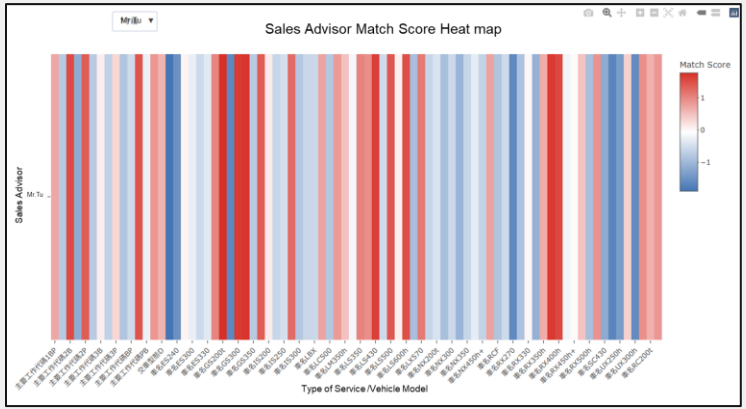
IPA：Explore diversity of every individuals

Translation of Findings:

“To identify every Sales advisor characteristics of service ,finding which topic improve priority.”

Strategic Insight:

As shown in my IPA Chart identified that “Service Duration Time Satisfaction" (item 32) , its performance was lagging behind the mean. This provided a clear "Concentrate Here" signal for Sales Advisor Service impovement.



SA–Customer Context Match Score：

Translation of Findings:

“identify distinct "Sweet Spots" and "Risk Zones" based on the Context Match Score. high Match Score correlates with customer perceptions of 'Attentive Service' and 'Professional Competence.”

Strategic Insight:

This Match Score Heatmap enables us to move from static experience-driven dispatching to dynamic data-driven workload allocation.

Thank you

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